

SOFTWARE DEVELOPMENT INTERNSHIP PROJECT

ROAD.NET

Human Connection Beyond Borders

Express Yourself. Discover People. Build Meaningful Connections.

1. PROJECT BACKGROUND

ROAD.NET is an evolution of an earlier project concept designed to create a trusted digital environment connecting people seeking companionship, relationships, experiences and services with people offering those experiences.

The original concept identified a central problem: cultural and social barriers can make it difficult for people to openly express their desire for companionship or find suitable people, while lack of trust between parties can make existing arrangements difficult. The proposed solution was an unbiased digital intermediary that brings both parties into a trusted environment.

The original project also introduced:

1. Subscriber accounts;
2. Client accounts;
3. Multi-person/agency accounts;
4. Profile creation;
5. Location;
6. Age;
7. Gender;
8. Marital status;
9. Services;
10. Photos/videos;
11. Biographies;
12. Availability;
13. Payment methods;
14. Booking;
15. Confirmation;
16. Introduction codes;
17. Reviews and comments;
18. Subscriptions;
19. Privacy controls.

The internship challenge is to re-engineer this original concept into a safe, modern, easy-to-use global human-connection platform.

2. THE NEW VISION

ROAD.NET should not become another “swipe left / swipe right” dating application.

The objective is to build a platform where people can discover meaningful connections based on intention, compatibility, geography, interests, lifestyle and shared experiences.

The platform should support:

Relationships

1. Marriage;
2. Serious relationships;
3. Getting to know someone;
4. Friendship;
5. Companionship.

Global Connections

1. Local matching;
2. National matching;
3. Regional matching;
4. Diaspora matching;
5. Cross-border matching;
6. Intercontinental matching.

Experiences

1. Shared activities;
2. Events;
3. Wellness;
4. Travel;
5. Cultural experiences;
6. Social experiences.

Community

1. Good moments;
2. Interests;
3. Stories;
4. Photos;
5. Activities;

6. Discovering people naturally.

3. THE CORE PROBLEM

The developer must design around this problem:

People can meet thousands of people online, but meeting the right person, understanding their intentions, establishing trust and developing a meaningful connection remains difficult.

Existing online dating platforms demonstrate demand for digital relationship discovery, but research also shows concerns around harassment, scams, unwanted contact and uncertainty about how algorithms actually produce matches.

ROAD.NET therefore focuses on:

Intent + Compatibility + Trust + Controlled Introduction + Experience.

4. WHO USES ROAD.NET?

The original project uses the terms **Subscriber** and **Client**.

For this modern version, the system should support three principal account types.

A. INDIVIDUAL

A person looking for:

1. Love;
2. Marriage;
3. Serious relationship;
4. Friendship;
5. Companionship;
6. International connection.

B. EXPERIENCE / SERVICE PROVIDER

A person or business offering legitimate experiences or services such as:

1. Wellness;
2. Massage;
3. Couple experiences;

4. Events;
5. Social activities;
6. Travel experiences;
7. Cultural experiences.

C. ORGANIZATION / AGENCY

An account capable of managing multiple approved profiles or service providers.

This preserves the original **multi-person account** concept.

5. AGE REQUIREMENT

ROAD.NET shall be an **18+ platform**.

The original project explicitly establishes that users under 18 are not permitted.

The developer must therefore implement an appropriate age gate during registration.

6. REGISTRATION

The registration process should be extremely simple.

Step 1 — Account Type

What brings you here?

1. Find a meaningful relationship
2. Find a marriage partner
3. Meet new people
4. Make international connections
5. Discover experiences
6. Offer experiences/services

Step 2 — Basic Profile

1. First name/display name;
2. Date of birth/age;
3. Country;
4. City/region;
5. Gender;
6. Languages;
7. Marital status;
8. Profession;

9. Short biography.

The original document already establishes many of these profile fields.

7. “WHAT ARE YOU LOOKING FOR?”

This becomes one of ROAD.NET's most important features.

The user selects:

My intention

1. Marriage;
2. Serious relationship;
3. Dating;
4. Friendship;
5. Companionship;
6. Long-distance relationship;
7. Diaspora connection;
8. Cultural exchange;
9. Shared experiences.

The user can select more than one.

8. GEOGRAPHIC CONNECTION

The user chooses:

Where do you want to connect?

LOCAL

People near me.

NATIONAL

People in my country.

REGIONAL

People in my region.

DIASPORA

People from my country living abroad.

CROSS-BORDER

People in another country.

INTERCONTINENTAL

People on another continent.

GLOBAL

Anywhere.

This is one of the major differences from a conventional local dating app.

9. DIASPORA MATCHING

Create a dedicated:

DIASPORA

section.

Example:

I live in Canada.

My origin is Rwanda.

I am interested in connecting with Rwandans.

Or:

I live in Rwanda.

I am interested in meeting Rwandans living abroad.

This should become a core ROAD.NET feature.

10. INTERCONTINENTAL MATCHING

The system should allow:

Africa ↔ Asia

Africa ↔ Europe

Africa ↔ North America

Africa ↔ South America

etc.

The user controls their geographical preferences.

11. PROFILE

A profile should contain:

About Me

Short biography.

My Intentions

What the person wants.

My Interests

Selectable categories.

My Lifestyle

Optional.

My Languages

Important for international matching.

My Location

Country and general location—not precise residential address.

My Photos

Authentic photographs.

Verification

1. Email verified;
2. Phone verified;
3. Other verification levels as the system evolves.

The original document requires authentic photos/videos and profile biographies.

12. COMPATIBILITY ENGINE

This is the heart of the internship.

The system should calculate a Compatibility Indicator.

Example:

87% Compatible

Why?

Relationship intention: 25/25
Geographic preference: 9/10
Interests: 13/15
Lifestyle: 12/15
Languages: 10/10
Long-distance preference: 5/5
Other preferences: 13/20

13. IMPORTANT: DO NOT CLAIM TO PREDICT LOVE

The developer must not build the system around the claim:

“Our AI knows who you will fall in love with.”

Research shows substantial skepticism about whether dating algorithms can actually predict love.

Instead:

“Our compatibility score shows how closely your stated preferences and interests align.”

This is transparent and explainable.

14. “WHY DID WE MATCH?”

Every recommended person should have:

Why you may connect

Example:

You both selected serious relationships, speak **English**, enjoy **travel**, and are open to long-distance relationships.

Then:

View Profile

Save

Introduce Yourself

This is one of ROAD.NET's signature features.

15. NO RANDOM SWIPING AS THE PRIMARY MECHANISM

ROAD.NET should not depend entirely on:

LIKE → DISLIKE → NEXT

Instead:

Discover

↓

Understand

↓

Compatibility

↓

Introduction



Mutual Acceptance



Connection

This creates a more intentional experience.

16. INTRODUCTION SYSTEM

Instead of allowing strangers to immediately obtain someone's private contact information, the platform should use:

Introduction Request

Example:

“Hello, I noticed that we both selected serious relationships and enjoy international travel. I'd like to get to know you.”

Recipient chooses:

Accept

Maybe Later

Decline

Only after acceptance should the next level of communication become available.

17. CONNECTION CODE

The original project uses an automatically generated **introduction code** after a successful transaction/connection.

Modern ROAD.NET can retain the concept but use it differently.

Every accepted connection receives:

Connection ID

Example:

RD-2026-8F24

This can be used for:

1. Support;
2. Reporting;
3. Booking;
4. Experience coordination;
5. Dispute resolution.

18. MOMENTS

Introduce:

MOMENTS

Users can share positive, appropriate content such as:

1. Travel;
2. Food;
3. Hobbies;
4. Culture;
5. Celebrations;
6. Achievements;
7. Places visited;
8. Activities;
9. Personal interests.

This allows people to discover each other through **what they enjoy**, rather than only through photographs.

19. SHARED MOMENT → CONNECTION

A user sees:

Sarah shared a moment from Kyoto.

Sarah writes:

“My first experience with Japanese tea culture.”

Another user can:

Like

Comment

React

View Profile

Send Introduction

This creates a natural bridge between social discovery and relationship discovery.

20. EXPERIENCE MARKETPLACE

This is where we preserve an important part of the original project.

Instead of limiting providers to relationship-oriented services, ROAD.NET should support legitimate experiences.

Examples:

Wellness

1. Massage;
2. Spa;
3. Relaxation.

Social

1. Events;
2. Dinner experiences;
3. Group activities.

Travel

1. Tours;
2. Local experiences;
3. Cultural activities.

Couples

1. Couple activities;
2. Private experiences;
3. Romantic experiences.

Providers can create listings with:

1. Description;
2. Photos;
3. Location;
4. Availability;
5. Price;
6. Duration;
7. Booking rules.

The original document already contains service selection, pricing, availability and booking functionality.

21. BOOKING SYSTEM

The original project already contains an important booking mechanism:

1. Select service.
2. Select time.
3. Make required payment.
4. Provider confirms.
5. Appointment becomes confirmed.
6. Time becomes unavailable.

The intern should implement a **cleaner and safer version** of this architecture.

22. AVAILABILITY CALENDAR

Providers should be able to configure:

1. Working days;
2. Working hours;
3. Available slots;
4. Duration;
5. Price;
6. Booking status.

Once a booking is confirmed:

The slot becomes unavailable.

This prevents double booking.

23. PAYMENT ARCHITECTURE

The original project identifies:

1. Bank payment;
2. Mobile money;
3. Card payment.

For the internship MVP, implement a payment simulation/mock gateway unless real payment credentials are available.

The architecture should nevertheless be designed so real payment gateways can later be integrated.

24. REVIEWS

After an experience or completed connection-related service, the user may be allowed to provide:

Rating

★★★★★

Review

The original document includes review/comment functionality.

Reviews should be moderated and connected to actual completed transactions where applicable.

25. SAFETY CENTER

This should be a mandatory feature, not an optional enhancement.

The platform should include:

Report User

Block User

Report Content

Unmatch

Safety Information

Contact Support

Research shows safety, unwanted contact, harassment and scams remain significant concerns for online dating users.

ROAD.NET should therefore make trust part of the product itself.

26. PRIVACY

The original project already contains a privacy policy concept covering collection, use, access, modification and deletion of user information.

The new system must implement privacy by design.

Users should control:

1. Profile visibility;
2. Search visibility;
3. Location visibility;
4. Who can send introductions;
5. Whether moments are public;
6. Whether their profile appears internationally.

Never expose:

1. Exact home address;
2. Private telephone number by default;
3. Payment credentials;
4. Sensitive account information.

27. ADMINISTRATION DASHBOARD

The administrator should be able to manage:

Users

1. Active;
2. Pending;
3. Suspended;
4. Verified.

Profiles

1. Review;
2. Approve;
3. Reject;
4. Suspend.

Reports

1. User reports;
2. Content reports;
3. Safety incidents.

Services

1. Providers;
2. Listings;
3. Prices;
4. Availability.

Bookings

1. Upcoming;
2. Completed;
3. Cancelled.

Payments

1. Transactions;
2. Pending;
3. Completed;
4. Refunds.

Analytics

1. Users;
2. Countries;
3. Matches;
4. Connections;
5. Bookings.

28. SUBSCRIPTION MODEL

The original document proposes free and paid subscription levels.

For the internship, the developer should build the architecture for:

FREE

Basic profile and discovery.

PLUS

Additional discovery and notification features.

PREMIUM

Advanced international matching and other premium functionality.

The exact prices do not need to be implemented during the internship.

The objective is to demonstrate that the software architecture can support subscription tiers.

29. NOTIFICATIONS

The system should notify users about:

1. Introduction requests;
2. Accepted introductions;
3. New compatible connections;
4. Comments;
5. Moments;
6. Bookings;
7. Payment status;
8. Verification status;
9. Safety alerts.

The original project already anticipates confirmation emails and dashboard notifications.

30. SEARCH

Users should be able to filter by:

1. Country;
2. Region;
3. Age range;
4. Gender;
5. Relationship intention;
6. Language;

7. Interests;
8. Diaspora;
9. Distance;
10. Long-distance preference;
11. Verified status.

The search must be easy enough for a first-time user to understand immediately.

31. DATABASE DESIGN

Minimum entities:

1. Users
2. Profiles
3. Preferences
4. Countries
5. Regions
6. Languages
7. Interests
8. Matches
9. Compatibility Scores
10. Introduction Requests
11. Connections
12. Moments
13. Comments
14. Reactions
15. Services
16. Providers
17. Availability
18. Bookings
19. Payments
20. Reviews
21. Reports
22. Blocks
23. Notifications
24. Subscriptions
25. Admin Users

The candidate must provide an ER diagram.

32. TECHNICAL ARCHITECTURE

The candidate may choose an appropriate stack.

Examples:

Frontend

React / Next.js / Vue / equivalent.

Backend

Laravel / Node.js / Django / equivalent.

Database

PostgreSQL / MySQL / equivalent.

Authentication

Secure authentication system.

Deployment

Candidate should explain how the system can be deployed.

The candidate is not required to use a specific technology if they can justify their choice.

33. API

The candidate should demonstrate appropriate API architecture.

Examples:

```
POST /register
POST /login
GET /profile
PUT /profile
GET /discover
GET /matches
GET /matches/:id
POST /introductions
GET /connections
POST /moments
POST /comments
POST /reports
POST /blocks
GET /services
POST /bookings
GET /notifications
```

The exact architecture is left to the developer.

34. SECURITY REQUIREMENTS

The candidate must demonstrate knowledge of:

1. Password hashing;
2. Authentication;
3. Authorization;
4. Input validation;
5. Secure APIs;
6. Protection against common web vulnerabilities;
7. Rate limiting;
8. Secure file uploads;
9. Access control;
10. Privacy;
11. Safe database practices.

35. CREATIVE REQUIREMENT

This is the most important part of the internship assessment.

The candidate must invent at least THREE mechanisms that distinguish ROAD.NET from ordinary dating applications.

For each:

Problem

What problem does it solve?

Mechanism

How does it work?

User Benefit

Why would users prefer it?

Technical Implementation

How would you build it?

36. EXAMPLES OF POSSIBLE INNOVATION

These are examples only. The candidate is encouraged to create better ideas.

A. CONNECTION JOURNEY

Discover → Introduction → Mutual Acceptance → Shared Experience → Connection.

B. WHY WE MATCHED

Explain compatibility instead of simply displaying a match.

C. MOMENT-BASED DISCOVERY

People can discover each other through interests and experiences.

D. DIASPORA BRIDGE

Connect people based on country of origin and current country.

E. CROSS-BORDER MODE

Users deliberately choose international matching.

F. INTRODUCTION REQUEST

No unsolicited private contact until mutual acceptance.

G. CONNECTION CIRCLE

Users can identify trusted people or communities who can help facilitate introductions in appropriate contexts.

The developer may propose completely different mechanisms.

37. WHAT THE INTERN MUST NOT DO

The applicant should not simply clone:

1. Tinder;
2. Bumble;
3. Badoo;
4. Hinge;
5. Match;
6. Airbnb.

The research shows that established dating platforms already provide large-scale digital matching, but also face concerns around safety, unwanted communication and uncertainty about algorithmic matching.

ROAD.NET must therefore have its own identity.

38. RESPONSIBLE DESIGN

The original document contains adult-oriented service concepts. For this software-development internship MVP, those elements should not be implemented as explicit sexual-service listings or sexual-service transactions.

Instead, the technical challenge should use safe, legitimate examples such as:

1. Wellness;
2. Massage;
3. Travel;
4. Cultural experiences;
5. Events;
6. Companionship;
7. Social experiences;
8. Relationship discovery.

This allows the intern to demonstrate the same marketplace, booking, payment, availability and matching architecture without making explicit sexual services the focus of the assessment.

39. REQUIRED DELIVERABLES

The candidate must submit:

1. Functional Web Application

A working prototype.

2. Source Code

Clean and organized.

3. Database

ER diagram + schema.

4. UI/UX

Wireframes or design system.

5. Matching Algorithm

Technical explanation.

6. API Documentation

Endpoints and authentication.

7. README

Including:

1. Project description;
2. Features;
3. Architecture;
4. Technology;
5. Installation;
6. Database;
7. Deployment;
8. Known limitations;
9. Future development.

8. Demonstration

The candidate must demonstrate:

Registration → Profile → Intent → Preferences → Discovery → Compatibility → Introduction → Connection → Moment → Experience → Booking → Review → Report.

40. MVP REQUIREMENT

The candidate does **not** need to build every future feature.

The minimum working product must demonstrate:

USER

- ✓ Registration
- ✓ Login
- ✓ Profile
- ✓ Intentions
- ✓ Preferences

MATCHING

- ✓ Search
- ✓ Discovery
- ✓ Compatibility
- ✓ Why-we-match explanation

CONNECTION

- ✓ Introduction request
- ✓ Accept/decline
- ✓ Connection

SOCIAL

- ✓ Moments
- ✓ Comments/reactions

EXPERIENCE

- ✓ Service listing
- ✓ Availability
- ✓ Booking

TRUST

- ✓ Verification concept
- ✓ Report
- ✓ Block

ADMIN

- ✓ Dashboard
- ✓ User management
- ✓ Report management

41. FUTURE ROADMAP

The developer should identify how the platform can later evolve into:

Phase 2

1. Mobile application;
2. Advanced matching;
3. Video introductions;
4. Translation;
5. Better verification;
6. International communities.

Phase 3

1. AI-assisted matching;
2. AI translation;
3. Recommendation engine;
4. Events;
5. Matchmaking communities;
6. Diaspora programs.

Phase 4

1. International payment infrastructure;
2. Verified experience marketplace;
3. Professional matchmaking;
4. Cross-border travel integration.

AI should assist discovery and recommendations—not claim to determine whether two people will fall in love. This is particularly important given the uncertainty users express about algorithmic predictions of love.

42. INTERNSHIP EVALUATION

Area	Points
Understanding of original ROAD.NET concept	10
Product thinking	10
UI/UX & simplicity	15
Frontend development	10
Backend development	10
Database architecture	10
Matching mechanism	10
Security & privacy	10
Creativity / innovation	10
Documentation & presentation	5
TOTAL	100

43. WHAT WE ARE REALLY TESTING

This is not simply a coding test.

We want to know whether the intern can:

Read an existing business concept → understand the problem → redesign the concept → architect the software → build an MVP → protect users → explain the system → propose its future.

That is much closer to the kind of software developer **ORIGIN GROUP** actually needs.

44. FINAL CHALLENGE TO THE CANDIDATE

Build ROAD.NET, a simple, secure and innovative human-connection platform that helps adults discover meaningful relationships, marriage opportunities, friendships, companionship, experiences and connections across communities, countries and continents.

The platform must support local, diaspora, cross-border and intercontinental discovery while allowing people to share positive moments, discover compatible people, introduce themselves respectfully and develop connections progressively.

Do not build another swipe-based dating application.

Build a platform around:

INTENTION + COMPATIBILITY + TRUST + EXPERIENCE + HUMAN CONNECTION.

The question behind the entire project:

“How can technology help two people who may never have met discover that they could meaningfully connect?”

That is the real software-development challenge.

